

Validere Data Consulting Interview Assignment

Please provide a data-driven solution to one of the following problems. Feel free to leverage any additional data sets or resources to achieve the depth of the solution that you are comfortable with. There is a preference for the data processing and analysis to be in Python, but we are flexible. We are evaluating your ability to:

- *Frame the problem*
- *Provide defensible assumptions*
- *Provide a candidate solution(s) to the problem*
- *Evaluate the solution(s) by leveraging data engineering and data science tools*
- *Evaluate the implications of the solution(s) on the business value*

There is no right answer to the questions, and the prompts have been left deliberately vague to test your ability to contextualize the problem. Please see the end of this document for the evaluation criteria.

1. Is there an ideal strategy to trade stocks as a mom-and-pop investor?

It is fairly common for people to invest and trade equity, whether this means trading stocks, indices, commodities, crypto assets, etc. Most proprietary traders have some sort of strategy they follow and many even build simple models to guide their trades.

Using any data available to you, propose a strategy to maximize profit for a trader for a mom-and-pop investor.

Part b) Evaluate the risk associated with following your proposed strategy. How can risk be quantified? How can risk be incorporated into the profitability of the strategy? How can the benefits and drawbacks of the strategy be articulated to a person with no data science background?

2. What qualities of crude oil lead to pipeline accidents?

Pipelines are generally regarded as a safe way to transport fuel, a far better alternative to tanker trucks or freight trains. But when a pipeline does fail, the consequences can be catastrophic; Since 1986, pipeline accidents have killed more than 500 people, injured over 4,000, and cost nearly seven billion dollars in property damage.

Using any resources available to you, can you determine what physical chemical properties of crude oil could be responsible for pipeline accidents?

Part b) What regulations on those properties would you recommend? Have we gotten better at regulating those properties over time? What implications on the oil market would these regulations have? How can success of these regulations be quantified in the short term and long term?

3. What is preventing widespread adoption of solar energy? Enough energy from the sun hits the earth every hour to power it for a year. Despite this, solar energy accounts for only a small fraction of total energy generated. Using any data you can find, can you determine what barriers prevent solar energy from being more widely adopted?

Part b) What are some policies or avenues through which it can become more ubiquitous? What are some potential downsides to these policies? What are the economic considerations of implementing these policies? How is it possible to quantify the success of these protocols, and how should the success metrics be defined?

4. Which North American energy companies stand to benefit the most from implementing a carbon emissions reduction strategy? Energy companies across the globe are being pushed - both by consumer sentiment and government regulation - to reduce their carbon footprint. The impact of carbon reduction is different for every company, as the associated costs and benefits will differ. Using information from public data sources (such as the [EPA emissions database](#) or the [Petrinex database](#)), evaluate the groups of companies that would benefit the most from implementing a carbon reduction program.

Part b) Are there any similarities between the companies you identified as benefiting the most? Are they generally motivated by regulations or consumer preferences? If you were at a company who helped implement carbon reduction strategies, what would your pitch be to these energy companies?

Skills Legend:

- **Critical Thinking:**
 - Were you able to frame the problem concisely and arrive at a solution that is defensible?
 - Are you aware of the consequences of the assumptions you are making?
 - Are you able to connect the solution to the business problem in the question?

- Were you able to quantify both the success criteria of your solution and the risk associated with the solution?
- **Software skills:**
 - Are you able to modularize your code by identifying the entities and operations on them?
 - Are you able to architect a software solution that is reliable, efficient, secure and minimizes complexity?
 - Are you able to configure the relevant ETL tooling to enable the bulk ingestion and analysis of data to support your conclusion?
- **Statistical/First Principles modeling:**
 - Can you see the underlying physical/economics etc phenomena driving the process?
 - Do you use the right statistical methods to support your hypotheses?